

Similarity by Selected Traits

Project: Transparent Book Recommender

1. Similarity by Selected Traits

Teach the learning cycle

Primary target: Represent items with numeric features

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf predictions from training examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one. Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

Concept to explain

Week 21: Similarity by Selected Traits

Term · Meaning

Term: similarity

Meaning: How alike two items are according to selected features and a declared comparison method.

Term: distance

Meaning: A specified measure of how far apart two representations are.

Term: coordinate

Meaning: A value identifying position along an agreed axis or within a space.

Term: scale

Meaning: The units and range used to represent or compare a quantity.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 22 as a two-session investigation using fictional data. A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 scale. A learner requests [2, 4]. We compare options by adding absolute differences. This is a chosen rule for similarity; it is not a learned model or an objective measure of quality. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

The absolute difference is 4.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can calculate a feature distance and state how scaling changes its meaning.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Reader target: length=2, difficulty=3.
2. Book A: (2,2) → grid distance $|2-2|+|2-3|=1$.
3. Book B: (4,3) → distance $|4-2|+|3-3|=2$.
4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as distance.

Exact explanation to model

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

Explain the added worked example

A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 scale. A learner requests [2, 4]. We compare options by adding absolute differences. This is a chosen rule for similarity; it is not a learned model or an objective measure of quality.

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both scales are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

Key terms

Term · Meaning

Term: Feature vector

Meaning: An ordered list of numeric features

Term: Distance

Meaning: A number describing difference under a chosen rule

Term: Manhattan distance

Meaning: The sum of absolute feature differences

Term: Normalization

Meaning: Rescaling features to a comparable scale

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Find the distance from target [2, 4] to [2, 4] and to [3, 3].

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Similarity by Selected Traits

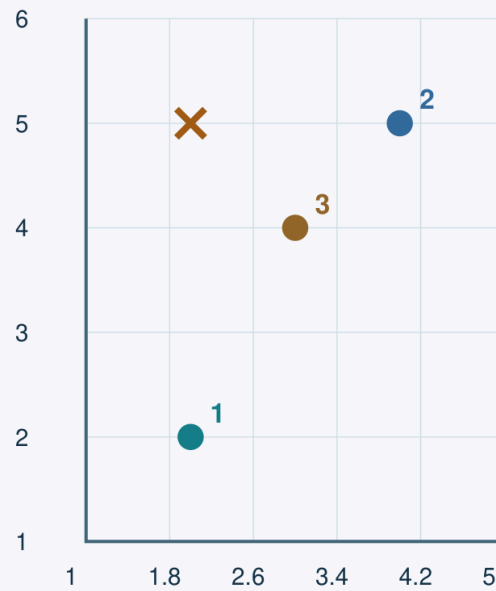
A fictional toy book matcher represents two preferences with numeric features.

The facts we will use

Target vector $[2,5]$. Candidate $A=[2,2]$, $B=[4,5]$, $C=[3,4]$.

Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance A

$$|2-2|+|5-2|=0+3=3.$$



2. Distance B

$$|2-4|+|5-5|=2+0=2.$$



3. Distance C

$$|2-3|+|5-4|=1+1=2.$$



4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Distance A

What happens and why: $|2-2|+|5-2|=0+3=3.$

Step: 2. Distance B

What happens and why: $|2-4|+|5-5|=2+0=2.$

Step: 3. Distance C

What happens and why: $|2-3|+|5-4|=1+1=2$.

Step: 4. Choose closest

What happens and why: B and C tie at distance 2; keep the tie visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Each coordinate contributes according to the declared distance rule. Smaller distance means closer in this representation, not automatically a better reading experience. If one coordinate uses a much larger numerical scale, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

What this example does not establish

A toy numeric encoding does not establish meaningful interpersonal distances or suitability of a recommendation.

Change one thing in the pictured case

Change: Give the first-coordinate difference weight 2 while the second keeps weight 1.

Prediction: Distances become $A=3$, $B=4$, $C=3$; A and C tie.

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Traits are [animals,adventure]. $A=[1,1]$, $B=[1,0]$, $C=[0,0]$; query= $[1,0]$.

Write the question to explain: Use number of differing traits as distance.

Reveal after predictions: Distances: $A=1$, $B=0$, $C=1$. B is nearest with equal feature weights.

Keep for an independent attempt: Score similarity using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Distances 0 and 2. Zero means identical recorded features, not identical activities in every way. Exit: No. Only the recorded features match; unrecorded needs and activity details may differ.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion task: From target [1, 2], find distances to A [2, 2] and B [0, 4].

Teacher check: A: $1 + 0 = 1$; B: $1 + 2 = 3$.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Distance between feature vectors. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Similarity depends on the features collected and the distance rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Explain why B ceases to be a closest option without any candidate data changing.

Reasoned teacher answer

Its first-coordinate gap 2 now contributes 4, raising its distance above A and C's 3.

Changed-case reasoning: Distances become $A=3$, $B=4$, $C=3$; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Plot six books on a 2-feature grid, compute simple grid distance, find nearest, then change one selected feature and compare.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Similarity depends on the features collected and the distance rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have distances 2, 1, 2. Y wins and X/Z tie.

A convenient shortcut preserves meaning: If one feature is duration in minutes up to 60 and another a rating up to 5, raw distance may be dominated by duration. Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Distances: $A=1$, $B=0$, $C=1$. B is nearest with equal feature weights.

Independent question

Score similarity using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

Expected independent answer

Each scores one point. The chosen traits create a tie; use a stated tie rule or request another relevant preference.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: "Similarity always gives exactly one winner." Keep the first attempt and an uncoached retry separate.

Original transfer answer

Equal-weight distances are $A=1$, $B=1$, $C=2$. Doubling the illustration mismatch makes $A=1$, $B=2$, $C=2$, so A becomes nearest. Weight choices express a task preference; they are not objective truth about book quality.

Explanations for the additional workbook examples

W1 Read a vector

For $[4, 1]$ under the worked feature order, identify energy and group-work ratings.

Energy 4, group work 1.

Use the declared feature order: position 0 means energy and position 1 means group work. The vector $[4,1]$ therefore records energy 4 and group work 1. Reversing the interpretation would change the meaning and could change the distance calculation.

W2 Compute distances

From target [1, 2], find distances to A [2, 2] and B [0, 4].

A: $1 + 0 = 1$; B: $1 + 2 = 3$.

W3 Repair feature order

One record uses [group work, energy] while others use [energy, group work]. Why is the distance unreliable?

It compares different meanings in each position. Reorder using the documented schema before computing.

W4 Rank independently

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

Distances A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

W5 Limit similarity

Two activities have distance zero. Must they suit every learner equally well?

No. Only the recorded features match; unrecorded needs and activity details may differ.

Assess independent transfer

Selected case: Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate distances, select the closest, and identify a tie among the others.

Selected case answer

Distances A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Transparent Book Recommender

Make selected traits and their scales visible in every recommendation explanation.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 22: Nearest-Neighbor Reasoning. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Use a small square grid and count horizontal plus vertical steps. Extension: rescale a minutes feature by its declared range and compare rankings transparently.

Independent W4 evidence: Distances A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule. Record whether the learner needed no help, a hint or explanation. Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have distances 2, 1, 2. Y wins and X/Z tie.

Next week: Weighted scores and constraints. Reuse this week's explicit contracts and evidence.

Plan the delayed return

Suggested return points in this level: ai-2/22, ai-2/24, ai-2/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive

project does not by itself demonstrate understanding.

Project evidence from this case

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

7. Prepare, teach and assess

Teaching target: Represent items with numeric features

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

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Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: From target $[1, 2]$, find distances to A $[2, 2]$ and B $[0, 4]$.

Reasoned practice answer: A: $1 + 0 = 1$; B: $1 + 2 = 3$.

Check every learner before discussion: Use the worked case for Distance between feature vectors. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Similarity depends on the features collected and the distance rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give the first-coordinate difference weight 2 while the second keeps weight 1. Require a prediction and a reason before comparing. Expected result: Distances become $A=3$, $B=4$, $C=3$; A and C tie. Reason: The representation's comparison rule changes; the supplied vectors stay the same.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.